

Binary Polarity Classifier : Sentiment Analysis

Satyam Kumar
Roll No: 2101187

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Introduction

- ▶ This project implements a binary polarity classification system using different machine learning and deep learning techniques.
- ▶ The dataset is obtained from the Cornell Movie Review Dataset.
- ▶ Different models were trained, evaluated, and compared, including:
 - ▶ Naive Bayes with Bag of Words (BoW)
 - ▶ Naive Bayes with TF-IDF
 - ▶ Support Vector Machine (SVM)
 - ▶ Deep Learning (Dense Neural Network)

Data Preprocessing

- ▶ Text preprocessing includes:
 - ▶ Lowercasing
 - ▶ Tokenization using NLTK
 - ▶ Removal of punctuation and stopwords
 - ▶ Stemming (Porter Stemmer)
- ▶ Text vectorization:
 - ▶ Bag of Words (BoW) using 'CountVectorizer'
 - ▶ TF-IDF using 'TfidfVectorizer'

Models Used

- ▶ Naive Bayes (MultinomialNB)
- ▶ Support Vector Machine (SVM)
- ▶ Deep Neural Network (Sequential Model)
 - ▶ Layers: Dense, Dropout
 - ▶ Optimizer: Adam
 - ▶ Loss: Binary Cross-Entropy
 - ▶ Regularization: L2

Model Comparisons

- ▶ Bag of Words with Naive Bayes achieved a decent performance but TF-IDF improved the results.
- ▶ SVM with TF-IDF achieved similar metrics to Naive Bayes, but slightly lower.
- ▶ Deep Neural Networks were regularized with dropout to improve generalization over the vanilla version.

Best Model Evaluation: Naive Bayes with TF-IDF

Set	TP	TN	FP	FN
Validation Set	392	383	117	108
Test Set	643	647	184	188

Metric	Validation Set	Test Set
Precision	0.77	0.78
Recall	0.78	0.77
F1-Score	0.78	0.78

Software Requirements

- ▶ Python 3.9+
- ▶ Jupyter Notebook
- ▶ Packages:
 - ▶ `scikit-learn`
 - ▶ `nltk`
 - ▶ `tensorflow`
 - ▶ `matplotlib`, `seaborn`
- ▶ Code download link: [GitHub Repository](#)

Execution Instructions

▶ Execution Instructions:

- ▶ Ensure all dependencies are installed.
- ▶ Clone the repository from GitHub.
- ▶ Navigate to the project directory.
- ▶ Open 'Assignment_final.ipynb' through Jupyter Notebook and select **Run All**.
- ▶ Results of all model outputs will be displayed.

Conclusion

- ▶ Binary polarity classification was successfully implemented using different models.
- ▶ Naive Bayes with TF-IDF achieved the best performance.
- ▶ Future improvements could involve experimenting with more advanced models like BERT.